AI 500 - Practice Problem Set 2

Q1: K-Means Clustering

You have been given the following 15 data points (A1-A15):

- A1 (2,10), A2 (2,6), A3 (11,11), A4 (6,9), A5 (6,4), A6 (1,2), A7 (5,10), A8 (4,9), A9 (10,12), A10 (7,5), A11 (9,11), A12 (4,6), A13 (3,10), A14 (3,8), A15 (6,11)

With K=3, run one iteration of K-means clustering. You are given the following centroids of the initial clusters:

1. Centroid 1=(2,6) is associated with cluster 1.
2. Centroid 2=(5,10) is associated with cluster 2.
3. Centroid 3=(6,11) is associated with cluster 3.

Answer the following:

a) Assign each point to its nearest centroid.
b) Compute updated centroids.

Q2: DBSCAN

Given the points A(3, 7), B(4, 6), C(5, 5), D(6, 4), E(7, 3), F(6, 2), G(7, 2) and H(8, 4), Find the core points and outliers using DBSCAN. Take ϵ = 2.5 and MinPts = 3.

Q3: K-Means

Apply the K**(=2)-Means** algorithm over the data (185, 72), (170, 56), (168, 60), (179,68), (182,72), (188,77) up to two iterations and show the clusters. Initially, choose the first two objects as initial centroids.

Q4: K-Medoid Clustering

You are given a cluster of 5 data points: A(2,6), B(3,7), C(4,6), D(3,5), E(8,2)

The initial medoid for this cluster is point **B(3,7)**. The cost is the sum of **Manhattan distances** from the medoid to every point in the cluster. (Manhattan distance = $|x1-x2|+|y1-y2|$).

Answer the following: (a) Calculate the total cost for the initial medoid, **B(3,7)**.

(b) The algorithm will now test swapping the medoid with a non-medoid point. Consider swapping with point **C(4,6)**. Calculate the new total cost if C were the medoid.

(c) Based on your calculations, should the swap be made? What is the new medoid for the cluster after this one update step?

Q5: DBSCAN

Consider the set of 2D data points plotted below. There are two dense, well-defined clusters and two outlier points.

Data Point Locations:

- **Cluster A:** 10 points centered around (3, 8)
- **Cluster B:** 10 points centered around (10, 3)
- **Outliers:** Point X(1,1) and Point Y(12,10)

You run DBSCAN on this data. Answer the following conceptual questions.

(a) With a well-chosen `epsilon` (ε) and `MinPts` = 3, how many clusters would the algorithm likely find? Which points would be classified as noise?

(b) What would likely happen if you keep `MinPts` = 3 but set the `epsilon` (ε) value to be very large (e.g., large enough to cover half the plot)?

(c) What would likely happen if you keep a reasonable `epsilon` (ε) but set `MinPts` to be very large (e.g., `MinPts` = 12)?